



wellcome
connecting
science

Genomic & Epidemiological Surveillance of Bacterial Pathogens

22-26 June 2025

Module 6: Web tools for Genomic
Epidemiology
Julio Diaz Caballero



Command Line



Web Tools

CONFIDENTIAL



Web Tools

Internal

What is Epidemiology?

- Epidemiology is the study of how often diseases occur in different groups of people and why.
- Epidemiological information is used to plan and evaluate strategies to prevent illness and as a guide to the management of patients in whom disease has already developed.

Epidemiology

John Snow & the Cholera Epidemic in Mid-19th Century London

ID	deaths	pumpID	distpump	distBSpump	BSpump	COORD_X	COORD_Y
1	3	1	88.048463737998915	88.048	1	529308.741420158185065	181031.351545980258379
2	2	1	84.400412550303329	84.400	1	529312.163571448763832	181025.172400684969034
3	1	1	82.319196917910901	82.319	1	529314.382429573452100	181020.293687972472981
4	1	1	79.916876055057088	79.917	1	529317.3796449454966	
5	4	1	77.809998314350267	77.810	1	529320.6754490596940	
6	2	1	62.833900718223070	62.834	1	529336.6841540620662	
7	2	1	106.425836511145988	106.426	1	529290.1467624248471	
8	2	1	95.623353231476500	95.623	1	529301.0228688720380	
9	3	2	109.814092391418285	111.679	0	529284.9986700989538	
10	2	2	107.006801780608157	108.423	0	529288.3593889662297	

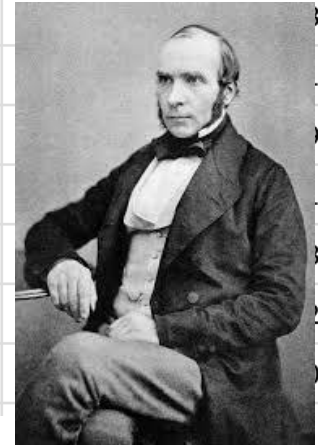


John Snow, 1854

Epidemiology

John Snow & the Cholera Epidemic in Mid-19th Century London

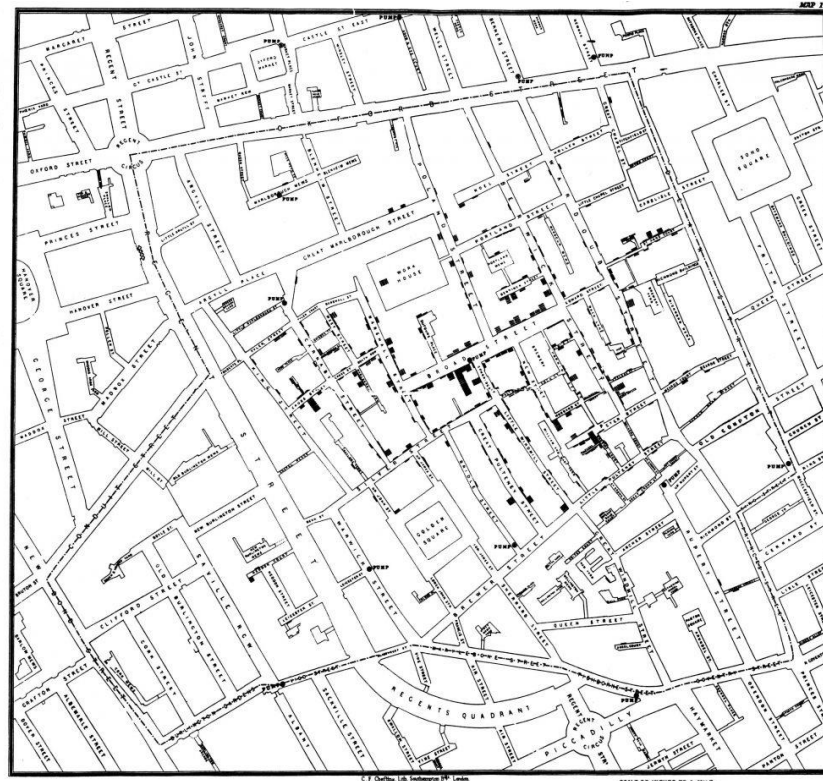
ID	deaths	pumpID	distpump	distBSpump	BSpump	COORD_X	COORD_Y
1	3	1	88.048463737998915	88.048	1	529308.741420158185065	181031.351545980258379
2	2	1	84.400412550303329	84.400	1	529312.163571448763832	181025.172400684969034
3	1	1	82.319196917910901	82.319	1	529314.382429573452100	181020.293687972472981
4	1	1	79.916876055057088	79.917	1	529317.379644945496693	181019.3975
5	4	1	77.809998314350267	77.810	1	529320.675449059694074	181025
6	2	1	62.833900718223070	62.834	1	529336.684154062066227	1810146
7	2	1	106.425836511145988	106.426	1	529290.146762424847111	1810184
8	2	1	95.623353231476500	95.623	1	529301.022868872038089	1810717
9	3	2	109.814092391418285	111.679	0	529284.998670098953880	1810162
10	2	2	107.006801780608157	108.423	0	529288.359388966229744	1810834



John Snow, 1854

Data Visualisation

John Snow & the Cholera Epidemic in Mid-19th Century London

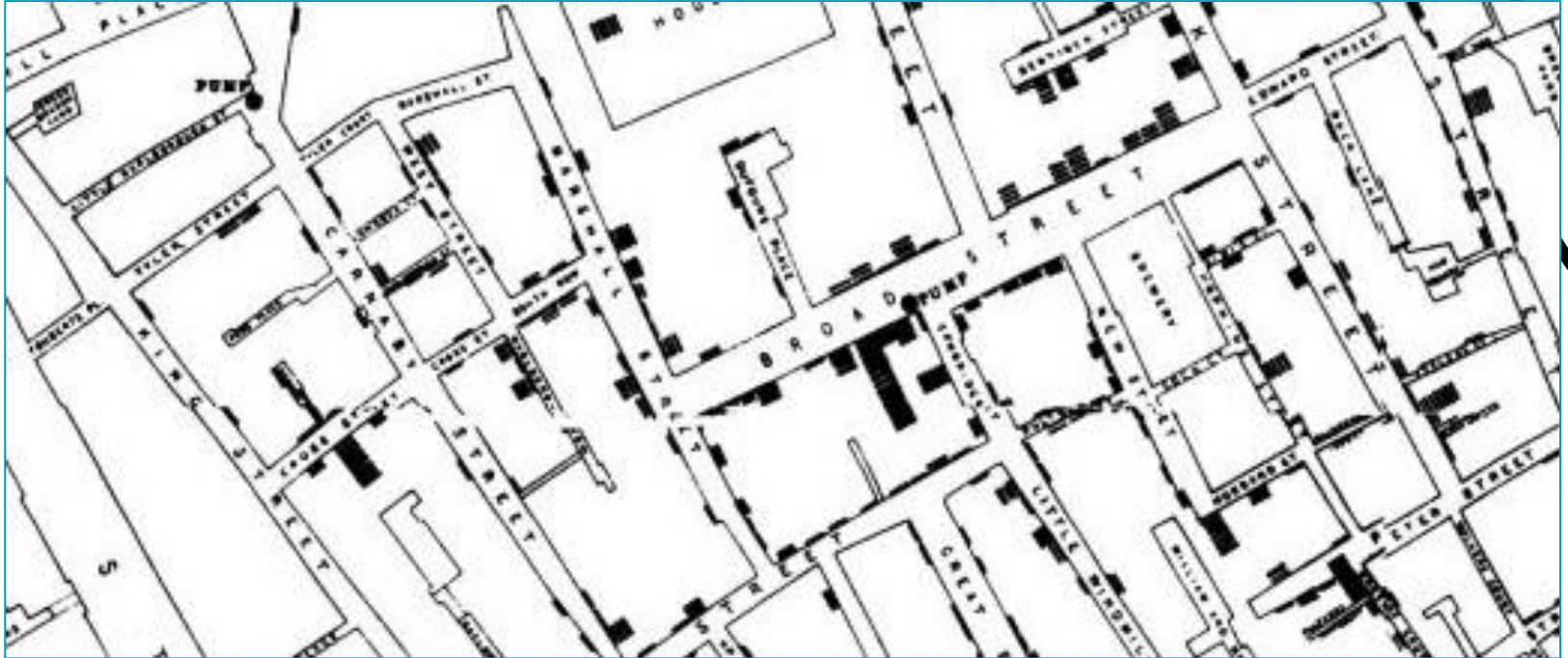


Internal

John Snow, 1854

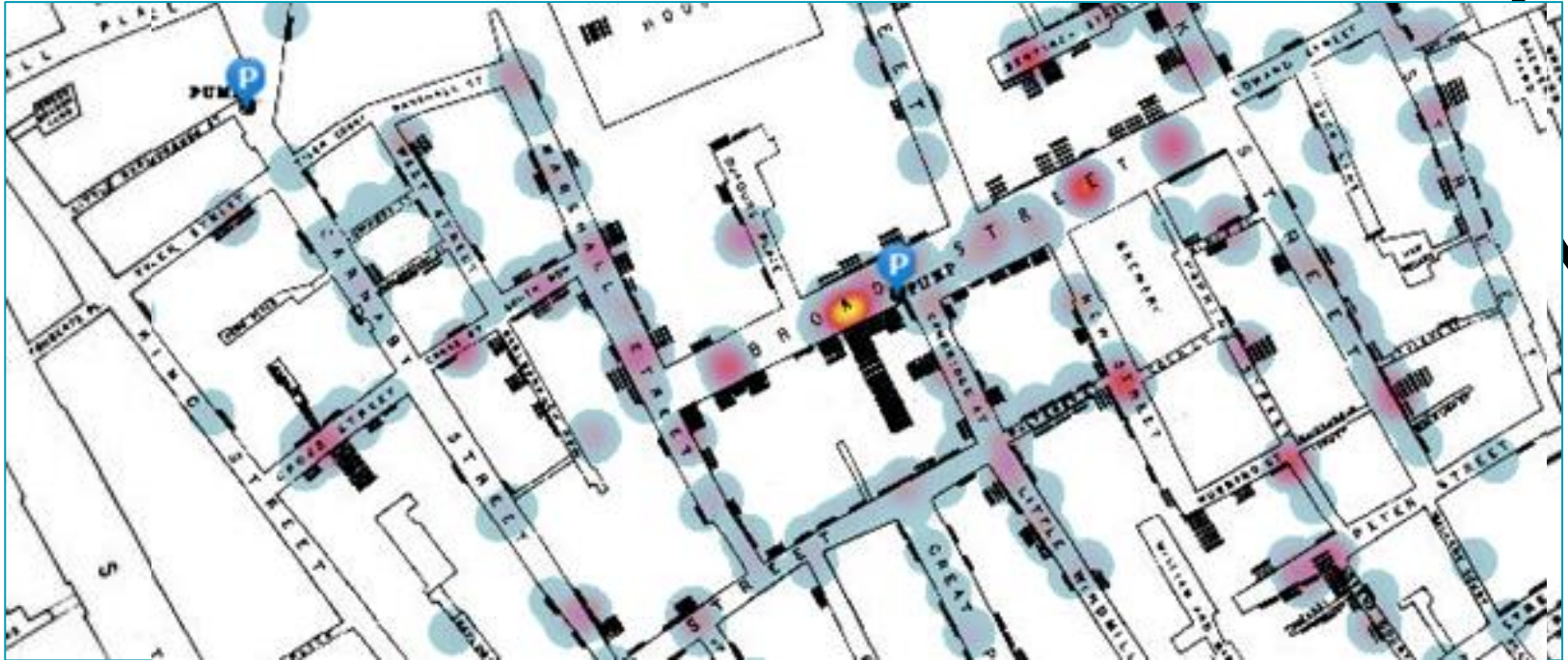
Actionable Insights

John Snow & the Cholera
Epidemic in Mid-19th Century
London



Actionable Insights

John Snow & the Cholera
Epidemic in Mid-19th Century
London



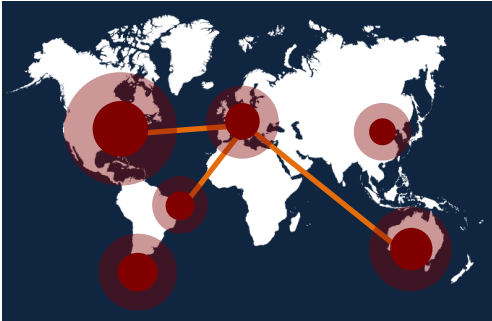
Today

John Snow & the Cholera Epidemic in Mid-19th Century London



Epidemiology in Space and Time

Global



International



National



Local



Epidemiology in Space and Time

Global



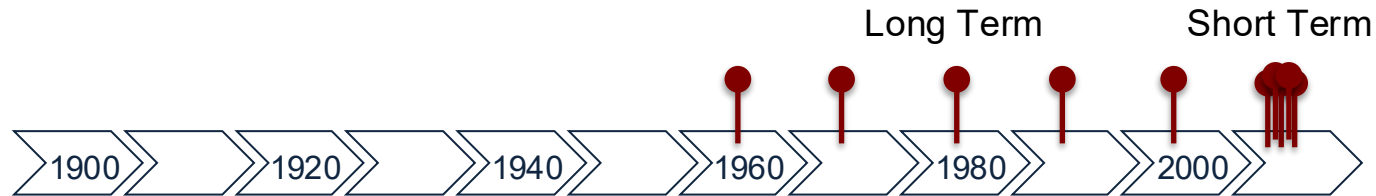
International



National



Local



Epidemiology in Space and Time

Local or Short-Term

- Is it a hospital outbreak caused by a single strain or independent infections caused by unrelated strains?
- What is the source of an outbreak?
- Are the strains resistant to antibiotics?

Global or Long-Term

- What is the worldwide origin of the prevalent lineage of a pathogen?
- Was the disease acquired by a patient during travel?
- When did resistance to an antibiotic emerge in a pathogen species?

Epidemiology in Space and Time – **WHERE?**

Local or Short-Term

- Is it a hospital outbreak caused by a single strain or independent infections caused by unrelated strains?
- What is the **source** of an outbreak?
- Are the strains resistant to antibiotics?

Global or Long-Term

- What is the worldwide **origin** of the prevalent lineage of a pathogen?
- Was the disease acquired by a patient during **travel**?
- When did resistance to an antibiotic emerge in a pathogen species?

Epidemiology in Space and Time – **WHEN?**

Local or Short-Term

- Is it a hospital outbreak caused by a single strain or independent infections caused by unrelated strains over **time**?
- What is the source of an outbreak?
- Are the strains resistant to antibiotics?

Global or Long-Term

- What is the worldwide origin of the prevalent lineage of a pathogen?
- Was the disease acquired by a patient during travel?
- **When** did resistance to an antibiotic emerge in a pathogen species?

Epidemiology in Space and Time – WHAT?

Local or Short-Term

- Is it a hospital outbreak caused by a **single strain or independent infections** caused by unrelated strains?
- What is the source of an outbreak?
- Are the strains resistant to antibiotics?

Global or Long-Term

- What is the worldwide origin of the prevalent **lineage of a pathogen**?
- Was the disease acquired by a patient during travel?
- When did resistance to an antibiotic emerge in a pathogen species?

Epidemiology in Space and Time – HOW?

Local or Short-Term

- Is it a hospital outbreak caused by a single strain or independent infections caused by unrelated strains?
- What is the source of an outbreak?
- Are the strains **resistant to antibiotics**?

Global or Long-Term

- What is the worldwide origin of the prevalent lineage of a pathogen?
- Was the disease acquired by a patient during travel?
- When did **resistance to an antibiotic** emerge in a pathogen species?

Information for Pathogen Surveillance

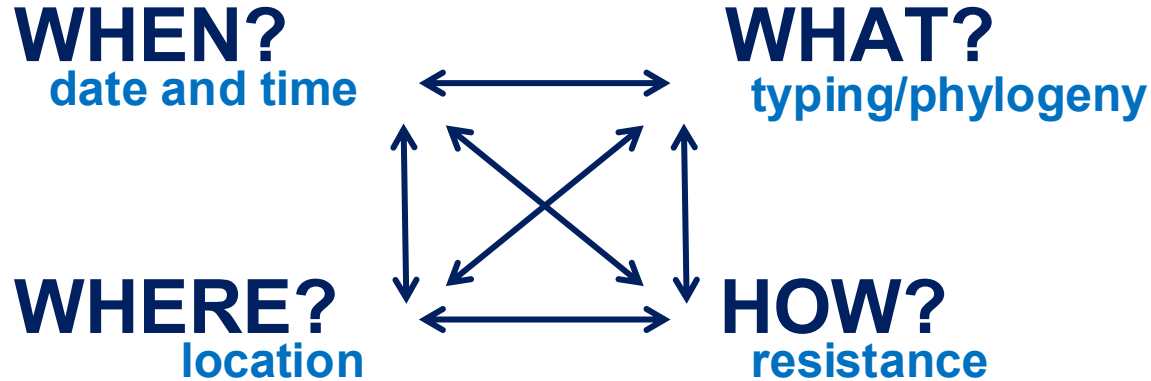
WHEN?
date and time

WHAT?
typing/phylogeny

WHERE?
location

HOW?
resistance

How do we bring data together?



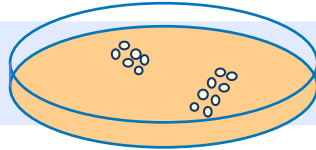
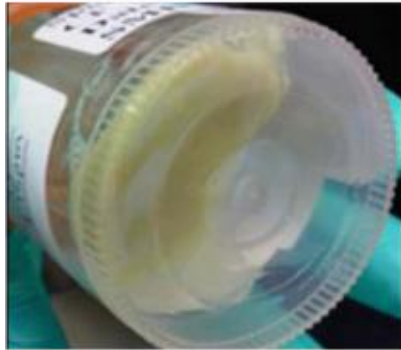
What is Genomic Epidemiology?

- Genomic epidemiology studies pathogen genetic variation across hosts, places, and time, integrating genomics with epidemiology.
- Sequencing and comparing pathogen genomes lets scientists map transmission, detect risk markers, and inform public action

Why do we use it in Public Health?

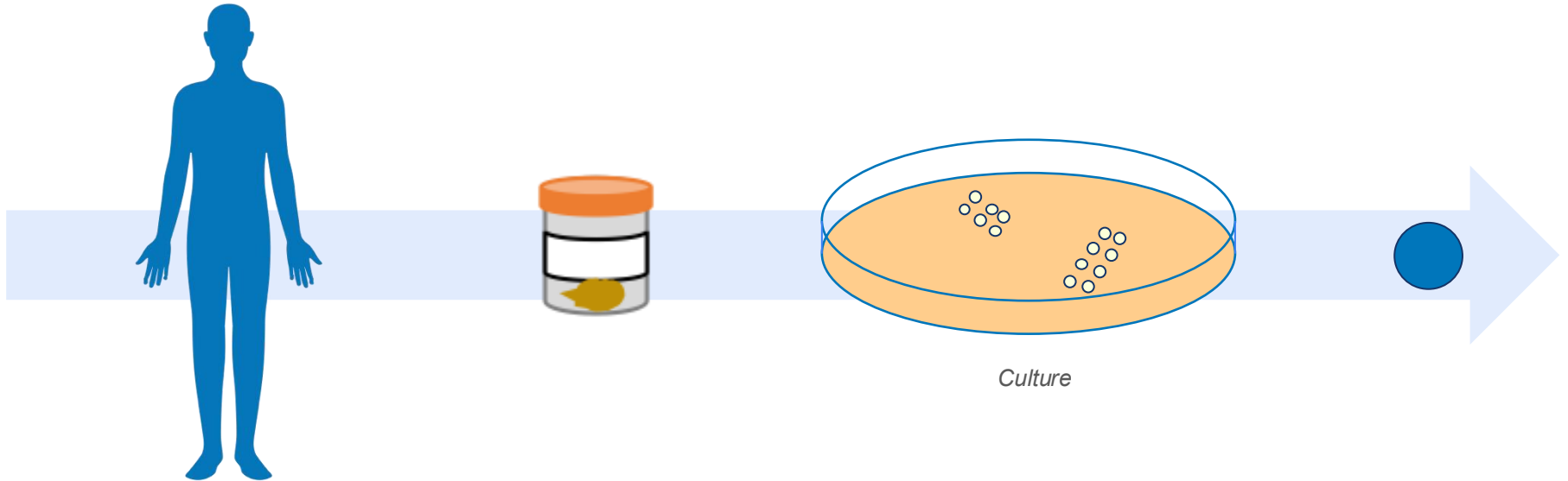
- Tracking outbreaks and identifying sources of recurrent infections
- Monitor antimicrobial effectiveness
- Vaccine development or modification
- Monitor policy changes
- New diagnostics
- Detect emergent threats

Next Generation Sequencing



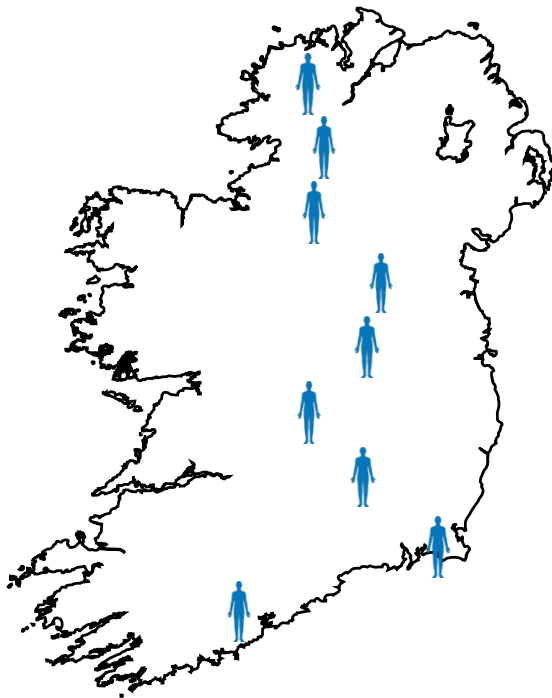
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ACACGATATGCAG
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Genomic Epidemiology Method

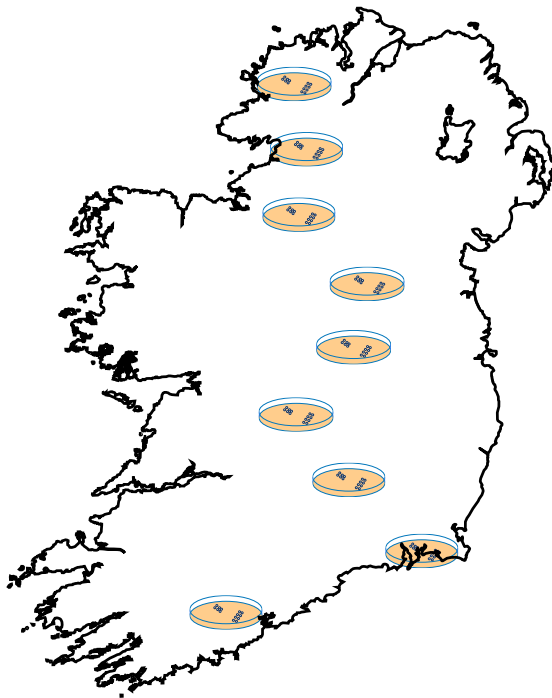


● Whole Genome of Single Isolate

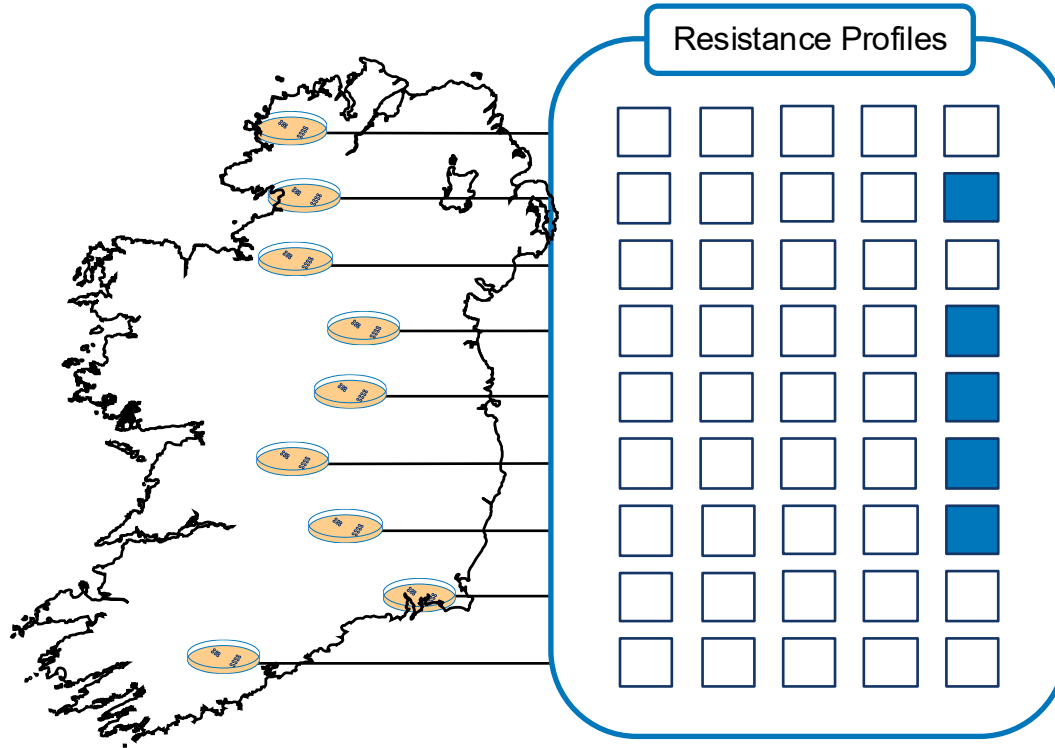
Genomic Epidemiology Method



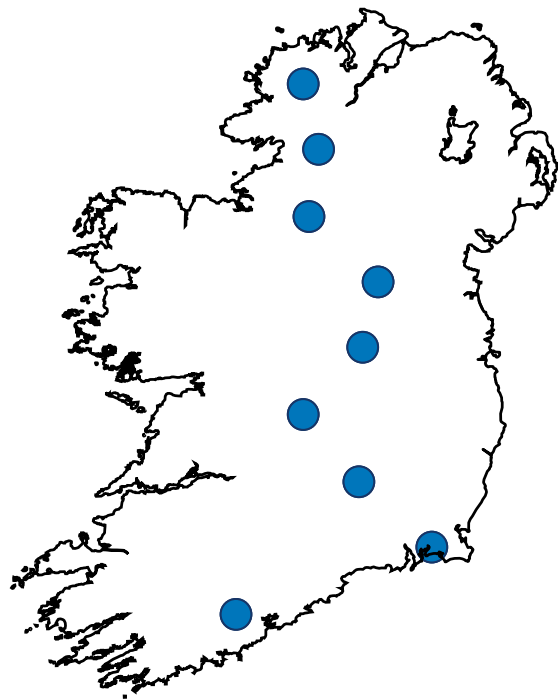
Genomic Epidemiology Method



Genomic Epidemiology Method

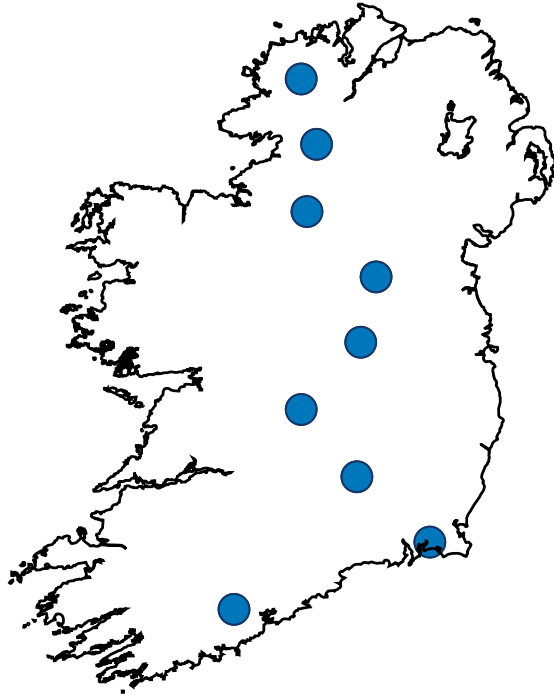


Genomic Epidemiology Method



● Whole Genome of Single Isolate

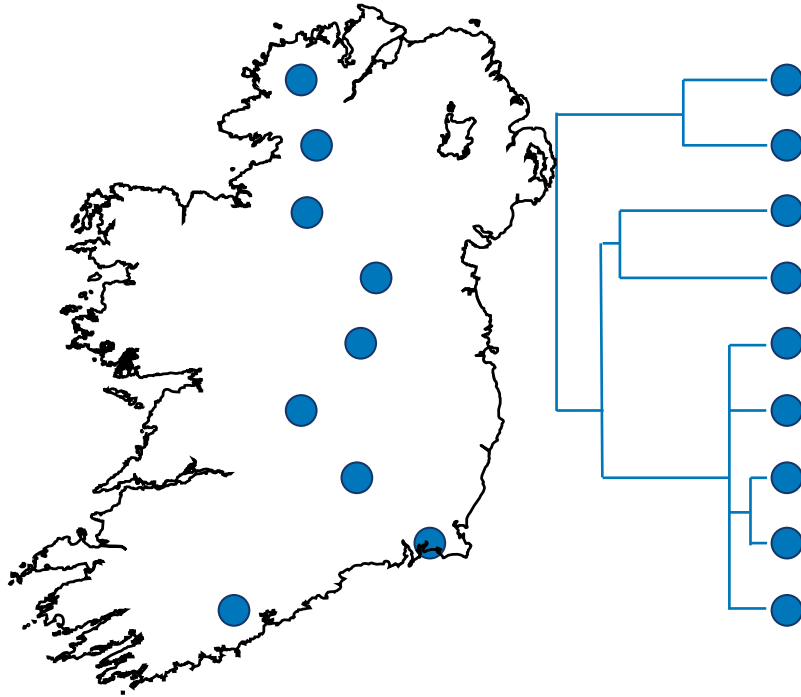
Genomic Epidemiology Method



Bioinformatics applications

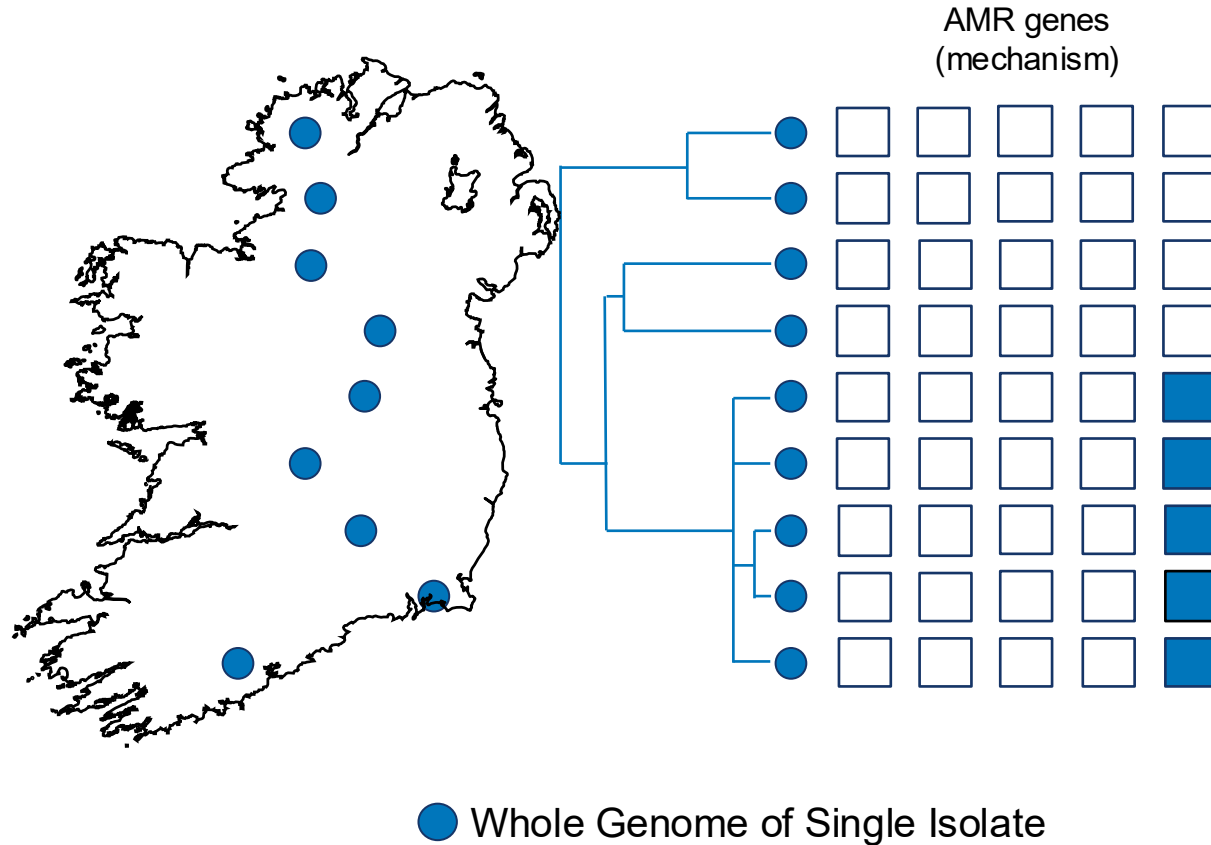
● Whole Genome of Single Isolate

Genomic Epidemiology Method

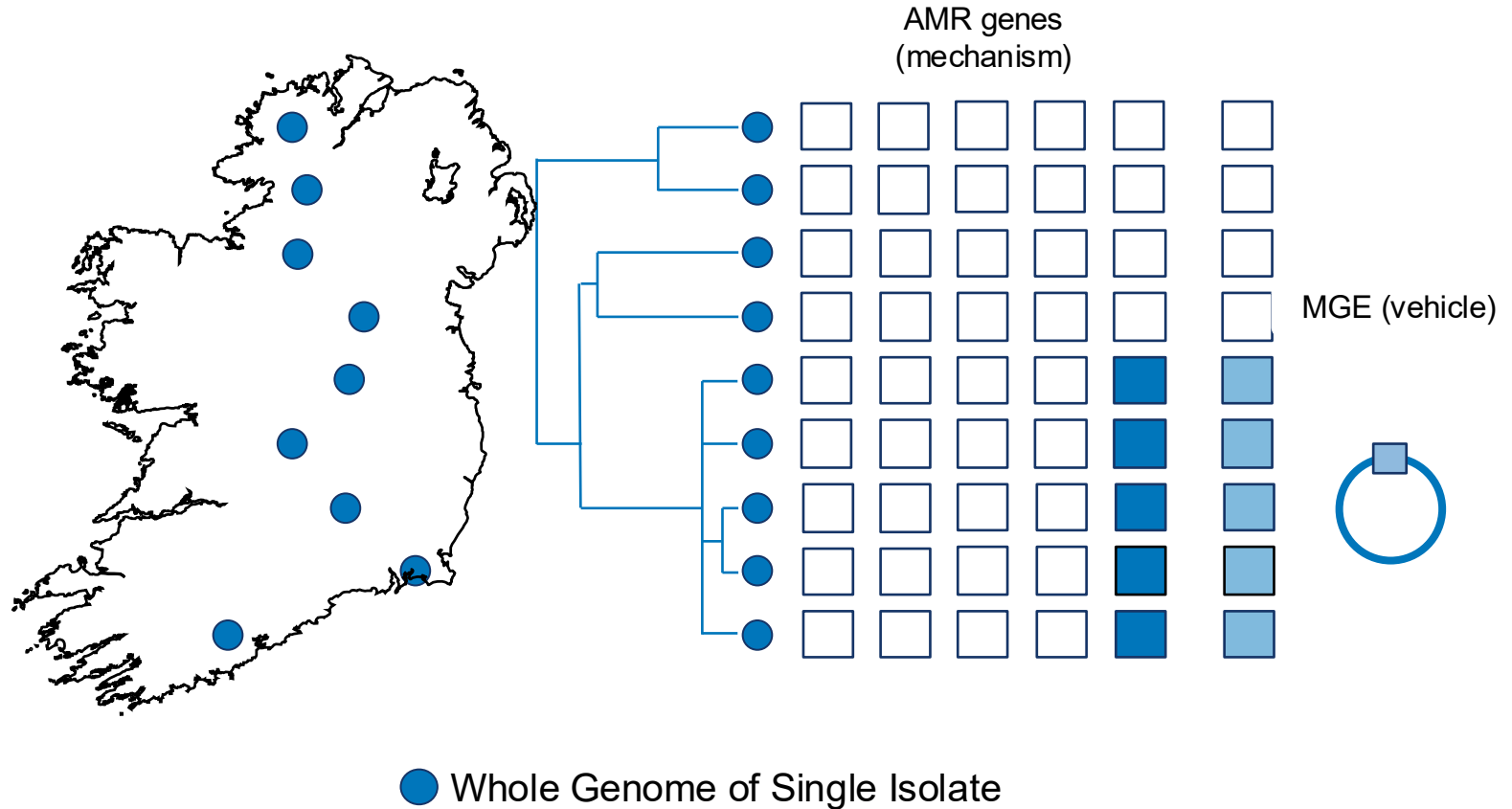


● Whole Genome of Single Isolate

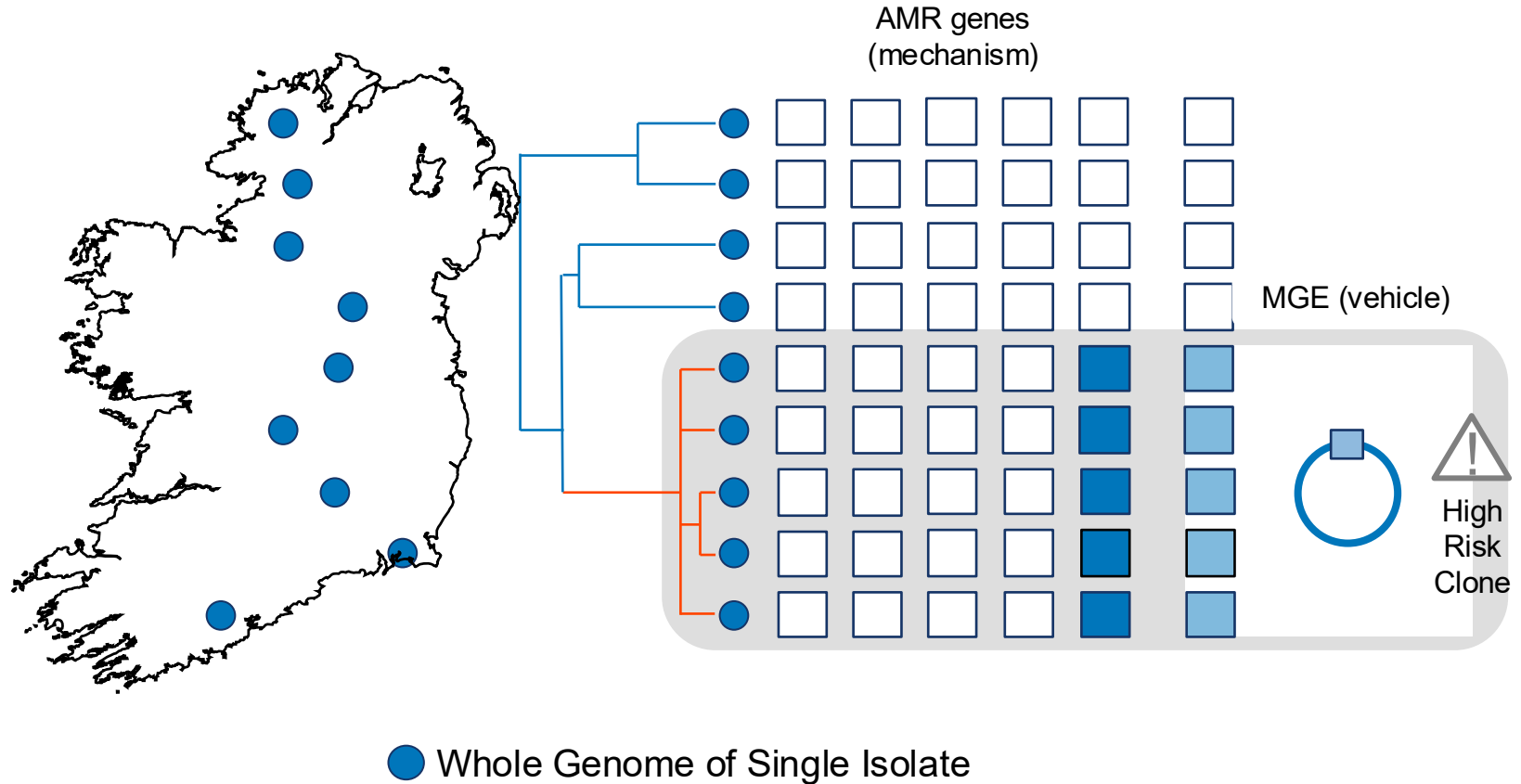
Genomic Epidemiology Method



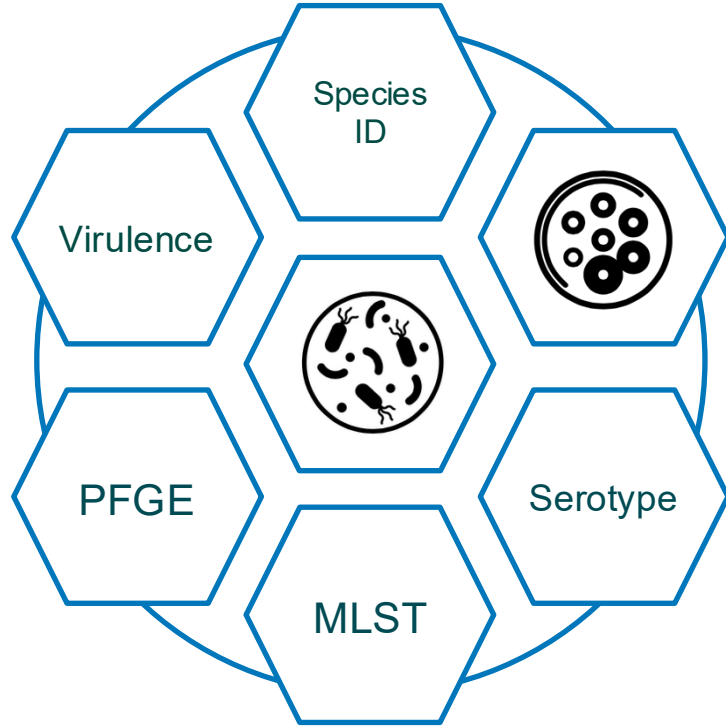
Genomic Epidemiology Method



Genomic Epidemiology Method



Extracting Epidemiological and Risk Markers from Genomes



WGS



Bix

[illegible][illegible]

Artemis File Entries Select View Goto Edit Create Run Graph Display

Entry: MAL1.emb1

1 selected bases on forward strand: 49715..49744

GC Content GC Windows size: 100

User algorithm: from MAL1.1 timepoints plot_window size: 16

Gene

Protein

Feature

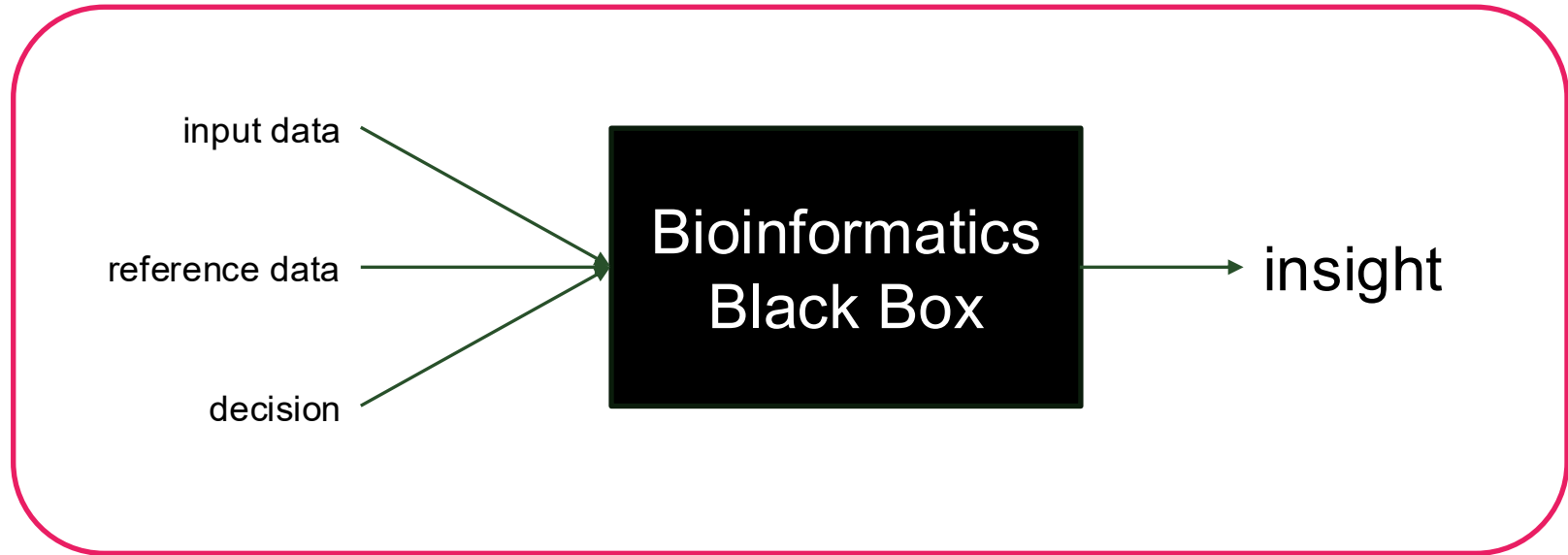
Legend

```

#SD      VN:1.0 50:coordinate
#SQ      SN:chr28  LN:64444167
#PG      ID:Tophat  VN:2.0.14  CL:/srv/data/tools/tophat/tophat -N 3 --read-dist 5 --read-len
            edit-dist 2 -i 50 -m 5000 -max-coverage-intron 5000 -m -out /data/user446/mapping/tophat/chr
            20 /data/user446/mapping/tophat/L6.18 GTAGAA_L087.R1.081.fastq
HWI-ST145:7.4:C1BDACXX:7:1102:4284:73714 1  chr28  1993930  3  100M  *  0  0
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      C  BBBCDDCCDDDDGDDDDDDDDDDCCCBBC70DDDDDDDDDDDDDDDDDDCCCGGDDCCCEDDBC70DDDDDDDDDDDDDDDDDDDDDDHHBFFDFFC@
AS1:-15  NM:i:3  XG:i:10  XG:i:10  MD:Z:55C20C1A9 NM:i:1.3  NH:1.2  CC:Z:  CP:55352714  HI:-1.8
HWI-ST145:7.4:C1BDACXX:7:1114:2759:41961 1  chr28  1993550  5  100M  *  0  0
      TCGCTTGATCATCTGATTTGATGGCTTCGACTCAGAGAGACCTGCTTCCGCTGGGGACGGAGACTCTCAGTGTATCCCTCGATACAGGGGGATGACGAC
      G  DDGDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD
AS1:-16  NM:i:3  XG:i:10  XG:i:10  MD:Z:68G16T1873 NM:i:1.3  NH:1.1
HWI-ST145:7.4:C1BDACXX:7:1204:14760:4038 1  chr28  270877 50  100M  *  0  0
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      C  DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD
AS1:-11  NM:i:2  XG:i:10  XG:i:10  MD:Z:BA85G13  NM:i:1.2  NH:1.1
HWI-ST145:7.4:C1BDACXX:7:1210:1167:8699 0  chr28  271218 50  50M470N50M  *  0  0
      0  GTGGCTTCACACAGGAATGTTGAGATGACATCTCATGCTGGGGTGCACTGGGCTCCGAAGCAGACATCTCCAAATAGCATCTTCG

```


Bioinformatics

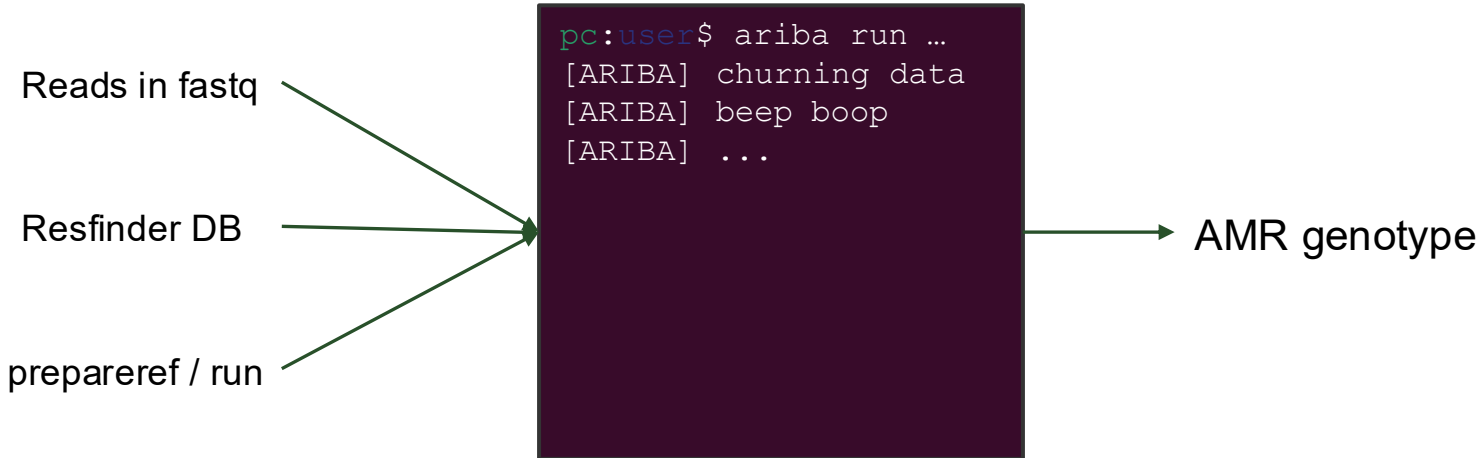


AMR Genotyping



AMR Genotyping

command line



AMR Genotyping

web-based

Sequences



The screenshot shows the Pathogenwatch website in a browser. The navigation bar includes links for GENOMES, COLLECTIONS, UPLOAD, and DOCUMENTATION. Below the navigation bar, there are tabs for 'NEW UPLOAD' and 'PREVIOUS UPLOADS'. The main heading asks 'What would you like to upload?'. There are three upload options:

- Single Genome FASTAs**: One or more FASTA files, one genome per FASTA file. (e.g. bacterial genomes). [Upload FASTA\(s\)](#)
- Multi-genome FASTAs**: Multiple genomes per file, one genome per record. (e.g. viral genomes). [Upload FASTA\(s\)](#)
- FASTQ**: One or more pairs of read files in FASTQ format. [Upload FASTQ\(s\)](#)



**AMR Genotype
and more**



So Julio, are web-based tools right for me?

Command Line Bioinformatics

pros

- Flexibility
- Customisability
- Articulation
- Availability
- No internet needed*
- Looking cool



Command Line Bioinformatics

pros

- Flexibility
- Customisability
- Articulation
- Availability
- No internet needed*
- Looking cool

cons

- Learning can seem daunting
- Requires adequate computing
- Installing software



Web-based Bioinformatics

pros

- Portable*
- No installations required
- No computational maintenance
- Easy to use
- Looking cool



Web-based Bioinformatics

pros

- Portable
- No installations required
- No computational maintenance
- Easy to use
- Looking cool

cons

- Difficult to customise
- Not all tools available*
- Requires internet*



The last mile problem

If a tree falls in a forest and no one hears it,
did it make a sound?

The last mile problem

If a discovery is made and nobody understands it, did it have impact?

The last mile problem: Napoleon’s Russian Campaign of 1812

Imperial French Army - Campaign Summary, December 1812

Memorandum regarding the strength of the Grande Armée during the Russian campaign.

Initial force crossing the Niemen River: 422,000 men.

Strength observed near Vilnius: 400,000 men.

Strength observed near Smolensk: 175,000 men.

Strength entering Moscow: 100,000 men.

Strength after crossing the Berezina: 28,000 men.

Estimated survivors reaching the Polish frontier: 10,000 men.

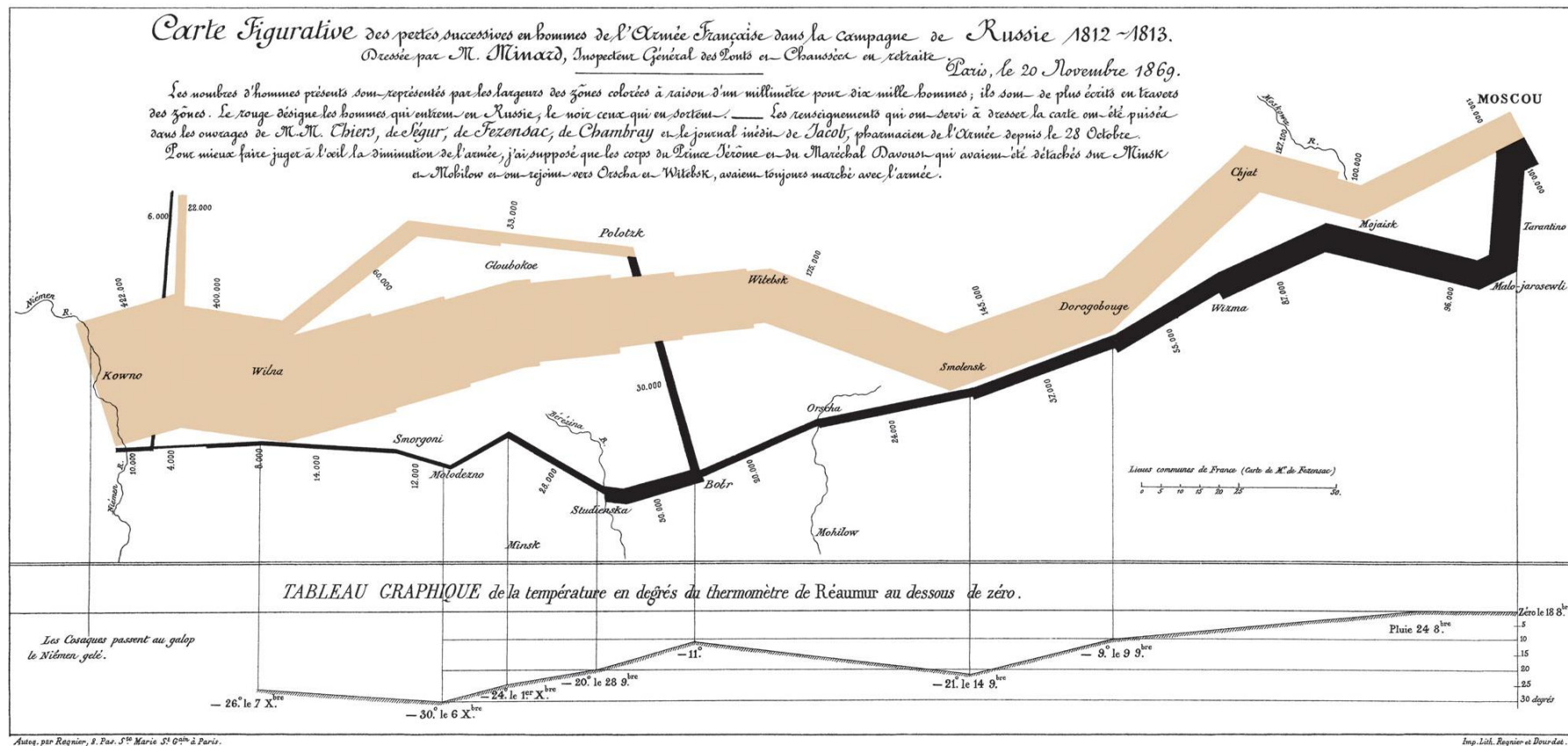
Losses attributed to disease, exhaustion, desertion, supply deficiencies, enemy action, and severe weather.

Appendix A: Troop Returns and Meteorological Observations

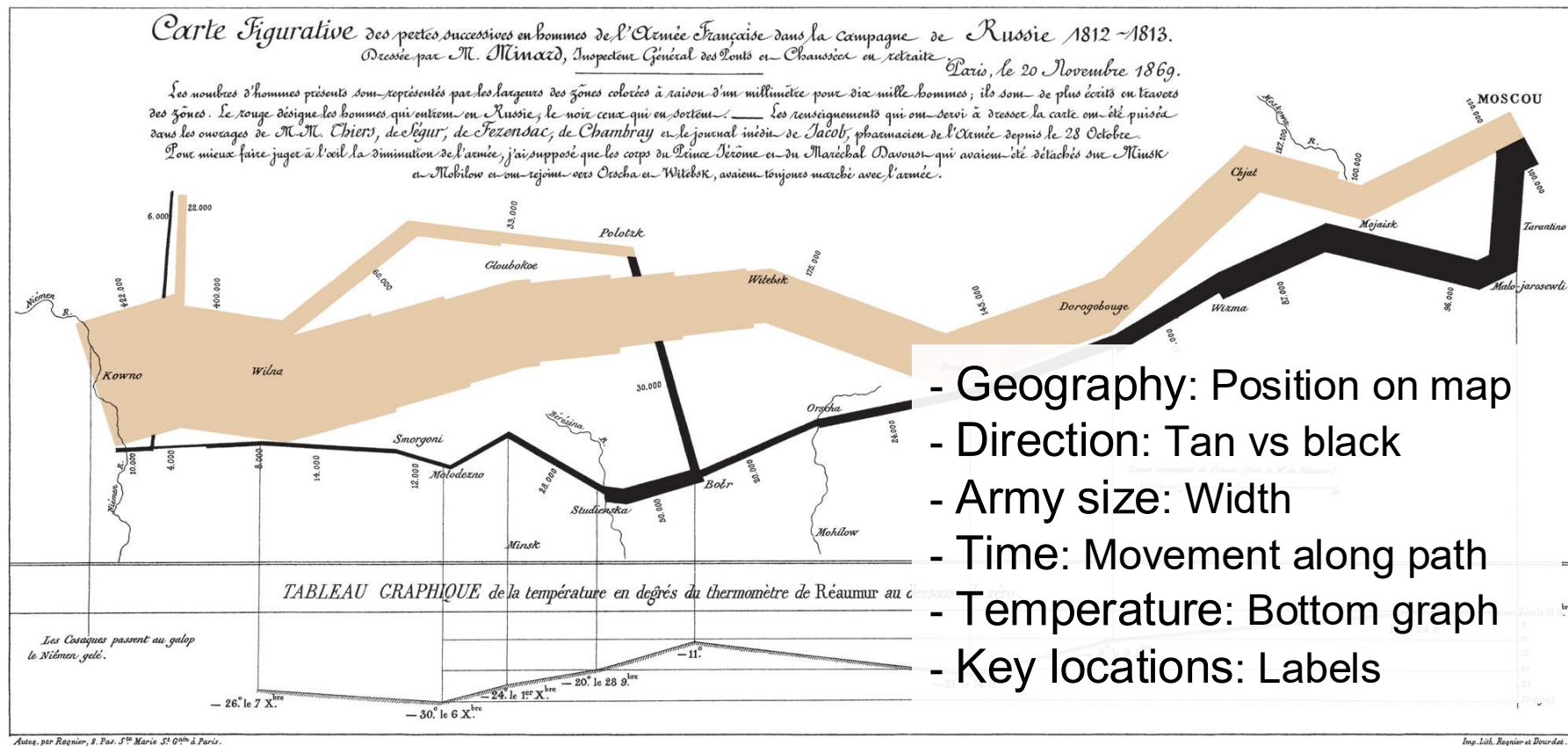
Date	Location	Troops Remaining
24 Jun	Niemen River	422,000
16 Aug	Smolensk	175,000
14 Sep	Moscow	100,000
28 Nov	Berezina	28,000
14 Dec	Frontier	10,000

Date	Location	Temp (°R)	Approx °C
14 Nov	Smorgoni	-9	-11
24 Nov	Molodechno	-21	-26
28 Nov	Berezina	-24	-30
6 Dec	Orsha	-30	-37.5

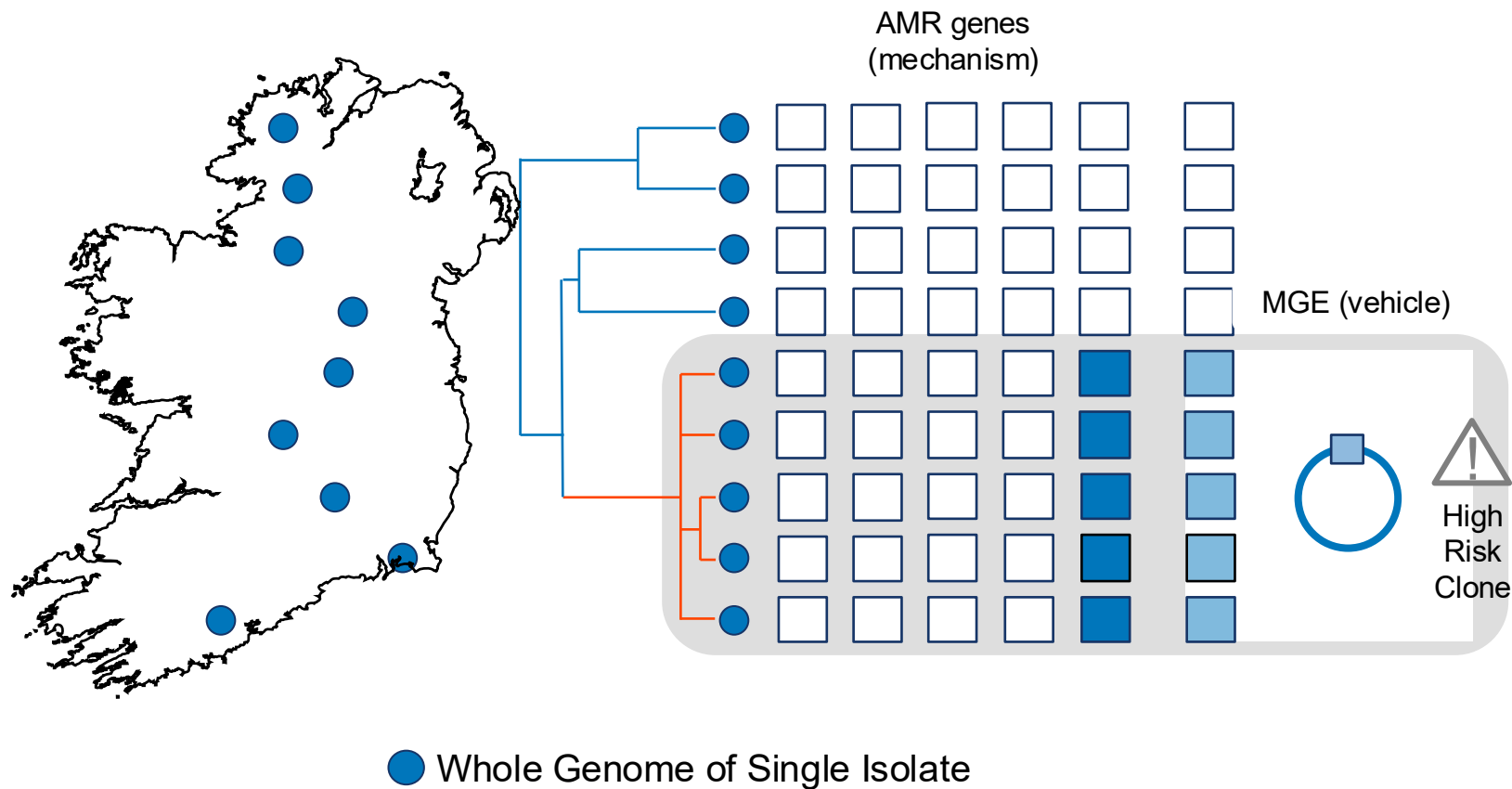
The last mile problem: Minard's Napoleon March



The last mile problem: Minard's Napoleon March



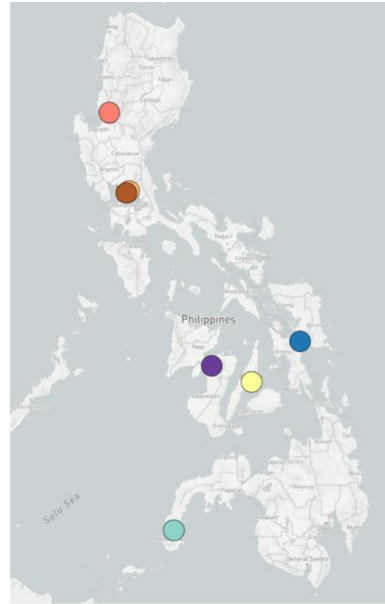
The last mile problem



National Antimicrobial Resistance Surveillance Program in the Philippines



Sonia Sia



National Antimicrobial Resistance Surveillance Program in the Philippines



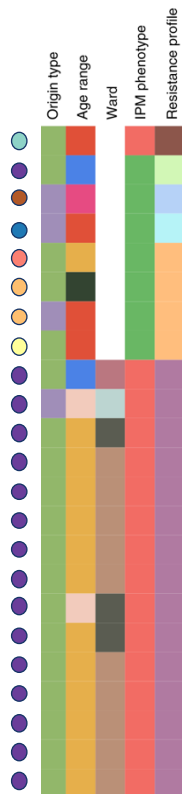
Sonia Sia



National Antimicrobial Resistance Surveillance Program in the Philippines



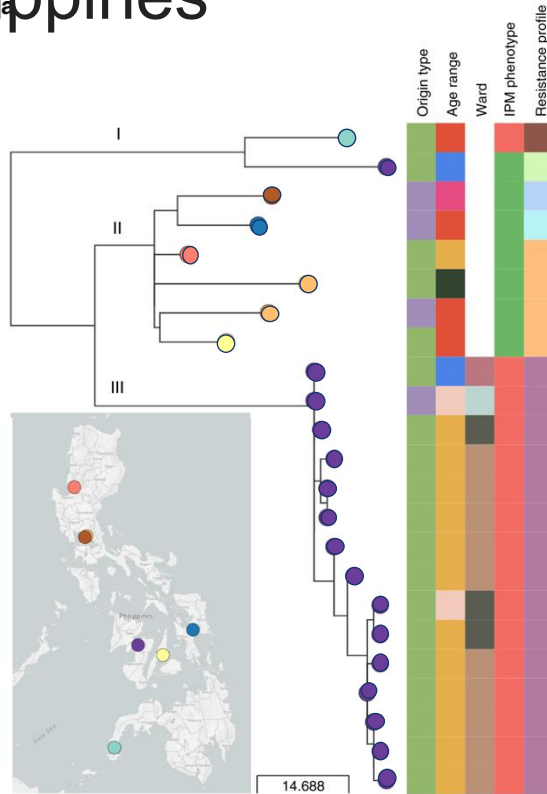
Sonia Sia



National Antimicrobial Resistance Surveillance Program in the Philippines



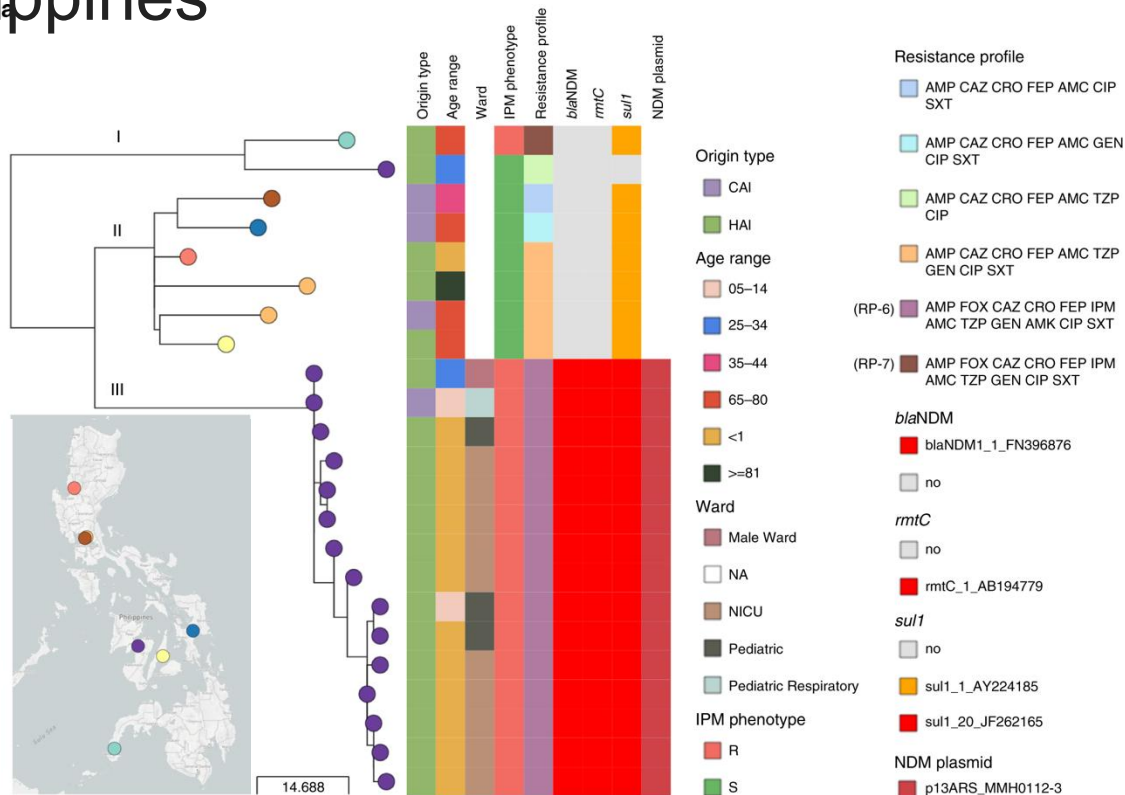
Sonia Sia



National Antimicrobial Resistance Surveillance Program in the Philippines

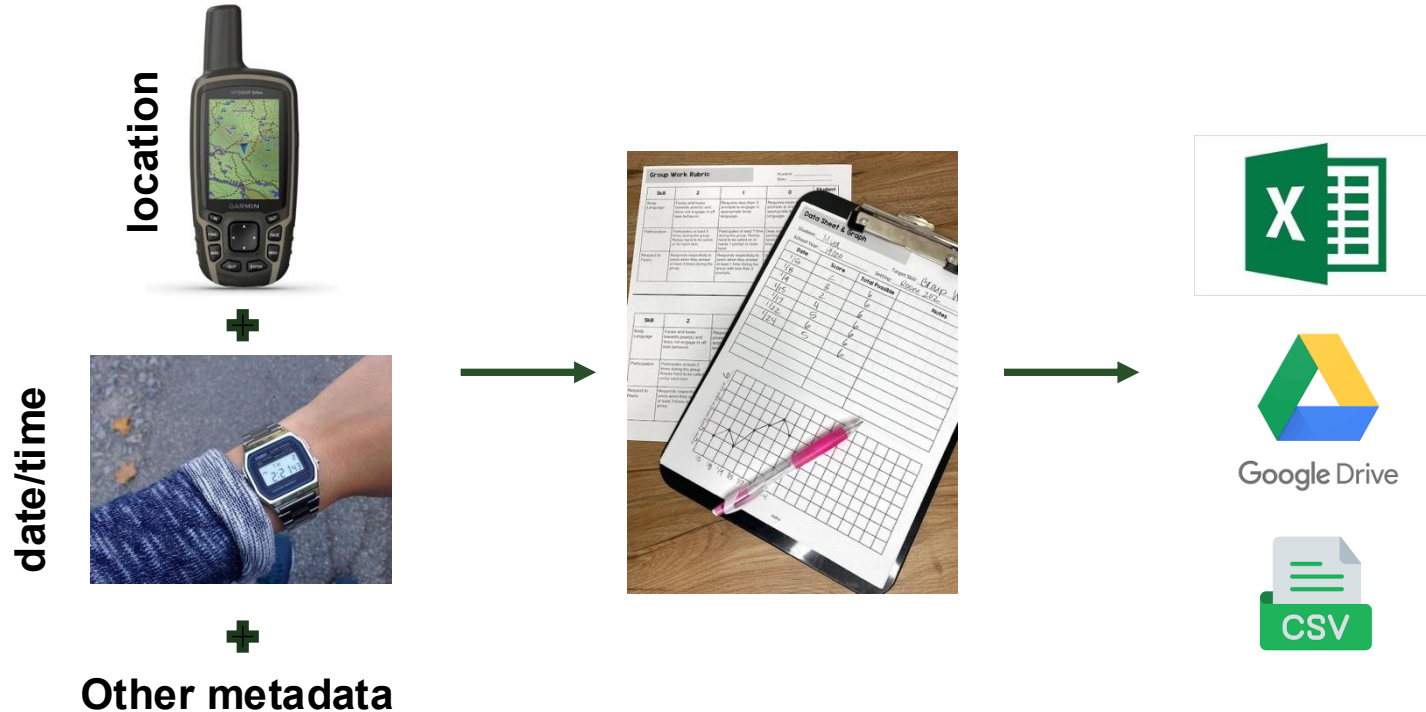


Sonia Sia



Genomic Epidemiology Challenges

DATA COLLECTION

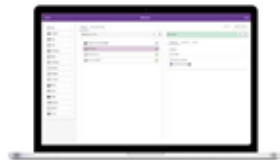




epi collect

<https://five.epicollect.net/>

- Web + Mobile based
- No specific data type
- No specific pathogen



Create



Collect



View



Internal

Project Samples



RURAL SURVEY 2018

Understanding the sentiments and thoughts of rural population in India

30591 entries

SOUTHERN REGION POLES

Low voltage
poles collection in Zimbabwe

76103 entries



Demo Time!



Genomic Epidemiology Challenges

DATA INTEGRATION

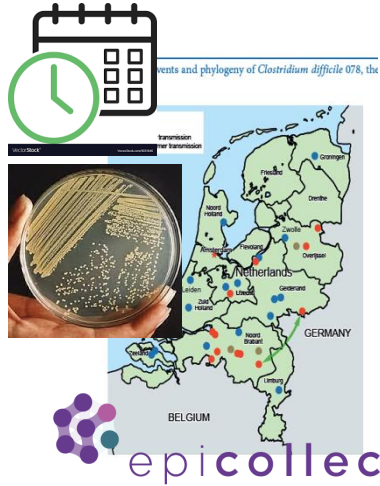


TABLE 1A

Clostridium difficile type 078 isolates used in this study, the Netherlands, 2002–11 (n=65)

R_L#P	Year	City	RT	Isolate	Source	Related Isolates	Association
8080_2#24	2006	Leiden	078	6072310	Clinic	Non-outbreak	Healthcare
8080_2#25	2006	Nijmegen	078	6086336	Clinic	Non-outbreak	Healthcare
8080_2#26	2007	Leiden	078	7001233	Clinic	Non-outbreak	Healthcare
8080_2#27	2007	Groningen	078	7004578	Clinic	Non-outbreak	Unknown
8080_2#28	2007	Utrecht	078	7005405	Clinic	Non-outbreak	Unknown
8080_2#29	2007	Zwolle	078	7021455	Clinic	Non-outbreak	Healthcare
8080_2#30	2007	Zwolle	078	7044912	Clinic	Non-outbreak	Community
8080_2#31	2007	Zwolle	078	7066827	Clinic	Non-outbreak	Community
8080_2#32	2007	Zwolle	078	7071308	Clinic	Non-outbreak	Healthcare
8080_2#33	2007	Zwolle	078	7086074	Clinic	Non-outbreak	Healthcare
8080_2#34	2007	Leiden	078	7091952	Clinic	Non-outbreak	Healthcare
8080_2#35	2008	Leiden	078	8011061	Clinic	Non-outbreak	Healthcare



Google Drive

idsheet

Insert Format Data Tools Add-ons Help All changes saved in Drive

	A	B	C	D	E	F
1	First name	Last name	State	City	Street Address	Join date
2	Ann	Cold	WI	Springfield	7 J Dr.	2/2/1990
3	Bob	New	NC	Sit	1411 Lincoln St.	7/10/2015
4	Ardo	Toothache	DE	Blot	787 An Tw	12/1/1980
5	JJ	Key	AK	Tilly	1 Kool-aid St.	11/25/2017
6	Po		7 HI	Boston	55 Globe	2/13/1990
7	III	Jeans	ND	Tub	1000 Broadway	8/8/2005
8	Gob 3.4	Poll	KY	Tab	2219 Lake Ave.	1/6/1963
9						
10						
11						
12						

Secret Public

TABLE 1

Results of Antimicrobial susceptibility testing

Isolate	Source	Strepto- mycin	Tn6235	Tetra- cycline	Tn6190
6072310	Clinic	Absent	Absent	Present	
6086336	Clinic	Absent	Absent	Present	
7001233	Clinic	NT	Absent	Absent	
7004578	Clinic	Absent	Absent	Present	
7005405	Clinic	Absent	Present	Absent	
7021455	Clinic	Absent	Absent	Present	
7044912	Clinic	Absent	Present	Absent	
7066827	Clinic	Absent	Absent	Present	
7071308	Clinic	Absent	Absent	Absent	
7086074	Clinic	Absent	Absent	Absent	
7091952	Clinic	Absent	Absent	Present	
8011061	Clinic	Absent	Absent	Absent	
8013820	Clinic	Absent	Absent	Absent	
805728	Clinic	Absent	Absent	Present	
8055344	Clinic	Present	Present	Absent	

Isolate	Source	Strepto- mycin	Tn6235	Tn6190
129280	Clinic	Absent	Absent	
H205	Farmer	Present	Present	
B37.3	Pig	Present	Present	Absent
53737	Clinic	NT	Absent	NT
47337	Clinic	Absent	Absent	Present
H102	Farmer	Absent	Absent	Absent
B31.3	Pig	Absent	Absent	Absent
B17.3	Pig	Absent	Absent	Absent
H121	Farmer	Absent	Absent	Present
B27.2	Pig	Absent	Absent	Absent
H230	Farmer	Absent	Absent	Present
H189	Farmer	Absent	Absent	Absent
B23.6	Pig	Absent	Absent	Present
H205	Farmer	Present	Present	Absent
B15.1	Pig	Absent	Absent	Absent



Genomic Epidemiology Challenges

DATA INTEGRATION



Google Drive

adsheet

Insert Format Data Tools Add-ons Help

All changes saved in Drive

	A	B	C	D	E	F
1	First name	Last name	State	City	Street Address	Join date
2	Alec	Cole	WA	Springfield	7 J Dr	2/1/1990
3	Bob	New	NC	St	1411 Lincoln St	7/10/2015
4	Arts	Todd	DE	Reh	787 W. Ter	12/1/1980
5	JJ	Kay	AK	Tal	1 Koonard St	11/20/2017
6	Pa		7 H	Brook	55 Glade	2/13/1990
7	H	Juan	ND	Tab	1000 Broadway	6/30/2005
8	Colt 3.4	Put	KY	Tab	2219 Lake Ave	1/6/1983
9						
10						
11						
12						

Secret Public



TABLE 1A
Clostridium difficile type 078 isolates used in this study, the Netherlands, 2002–11 (n=65)

R_LPT	Year	City	RT	Isolate	Source	Related Isolates	Association
8080_2#24	2006	Leliden	078	6072310	Clinic	Non-outbreak	Healthcare
8080_2#25	2006	Nijmegen	078	6086336	Clinic	Non-outbreak	Healthcare
8080_2#26	2007	Leliden	078	7001233	Clinic	Non-outbreak	Healthcare
8080_2#27	2007	Groningen	078	7004578	Clinic	Non-outbreak	Unknown
8080_2#28	2007	Utrecht	078	7005405	Clinic	Non-outbreak	Unknown
8080_2#29	2007	Zwolle	078	7021455	Clinic	Non-outbreak	Healthcare
8080_2#30	2007	Zwolle	078	7044912	Clinic	Non-outbreak	Community
8080_2#31	2007	Zwolle	078	7066827	Clinic	Non-outbreak	Community
8080_2#32	2007	Zwolle	078	7071308	Clinic	Non-outbreak	Healthcare
8080_2#33	2007	Zwolle	078	7086074	Clinic	Non-outbreak	Healthcare
8080_2#34	2007	Leliden	078	7095952	Clinic	Non-outbreak	Healthcare
8080_2#35	2008	Leliden	078	8011061	Clinic	Non-outbreak	Healthcare

TABLE 3
Results of Antimicrobial susceptibility testing

Isolate	Source	Strepto- mycin	Tn6235	Tetra- cycline	Tn6190
129280	Clinic	Absent	Present	Present	
H205	Farmer	Present	Present	Present	
B37.3	Pig	NT	Absent	Present	
7004578	Clinic	NT	Absent	Present	
7005405	Clinic	Present	Absent	Present	
7021455	Clinic	Present	Absent	Present	
7044912	Clinic	Present	Absent	Present	
7066827	Clinic	Present	Absent	Present	
7071308	Clinic	Present	Absent	Present	
7086074	Clinic	Present	Absent	Present	
7095952	Clinic	Present	Absent	Present	
8011061	Clinic	Present	Absent	Present	
8013820	Clinic	Present	Absent	Present	
805728	Clinic	Present	Absent	Present	
8055344	Clinic	Present	Present	Absent	

Isolate	Source	Strepto- mycin	Tn6235	Tn6190
129280	Clinic	Absent	Present	Present
H205	Farmer	Present	Present	Present
B37.3	Pig	NT	Absent	Present
7004578	Clinic	NT	Absent	Present
7005405	Clinic	Present	Absent	Present
7021455	Clinic	Present	Absent	Present
7044912	Clinic	Present	Absent	Present
7066827	Clinic	Present	Absent	Present
7071308	Clinic	Present	Absent	Present
7086074	Clinic	Present	Absent	Present
7095952	Clinic	Present	Absent	Present
8011061	Clinic	Present	Absent	Present
8013820	Clinic	Present	Absent	Present
805728	Clinic	Present	Absent	Present
8055344	Clinic	Present	Present	Absent



<http://data-flo.io>

- Web based
- Many input types (and growing)
- Many transformations (and growing)
- Many output types (and growing)

Demo Time!

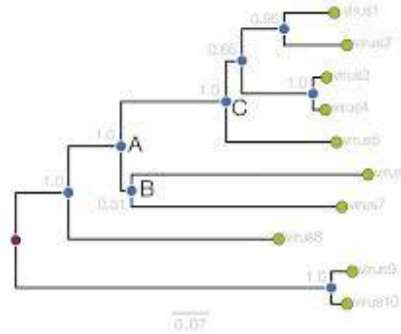


Genomic Epidemiology Challenges

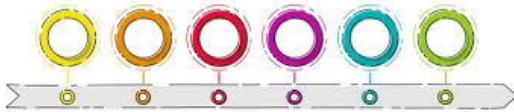
DATA EXPLORATION



map



phylogeny



timeline

GB-REFS Locid	GB-REFS Name	GB-REFS Title	GB-REFS No. Refs	GB-REFS TCF %	GB-REFS TCF #
VOL001	Paraguay001	Paraguay	128	507	142
VOL002	Paraguay002	Paraguay	128	507	142
VOL003	Paraguay003	Paraguay	128	507	142
VOL004	Paraguay004	Paraguay	128	507	142
VOL005	Paraguay005	Paraguay	128	507	142
VOL006	Paraguay006	Paraguay	128	507	142
VOL007	Paraguay007	Paraguay	128	507	142
VOL008	Paraguay008	Paraguay	128	507	142
VOL009	Paraguay009	Paraguay	128	507	142
VOL010	Paraguay010	Paraguay	128	507	142
VOL011	Paraguay011	Paraguay	128	507	142
VOL012	Paraguay012	Paraguay	128	507	142
VOL013	Paraguay013	Paraguay	128	507	142
VOL014	Paraguay014	Paraguay	128	507	142
VOL015	Paraguay015	Paraguay	128	507	142
VOL016	Paraguay016	Paraguay	128	507	142
VOL017	Paraguay017	Paraguay	128	507	142
VOL018	Paraguay018	Paraguay	128	507	142
VOL019	Paraguay019	Paraguay	128	507	142
VOL020	Paraguay020	Paraguay	128	507	142
VOL021	Paraguay021	Paraguay	128	507	142
VOL022	Paraguay022	Paraguay	128	507	142
VOL023	Paraguay023	Paraguay	128	507	142
VOL024	Paraguay024	Paraguay	128	507	142
VOL025	Paraguay025	Paraguay	128	507	142
VOL026	Paraguay026	Paraguay	128	507	142
VOL027	Paraguay027	Paraguay	128	507	142
VOL028	Paraguay028	Paraguay	128	507	142
VOL029	Paraguay029	Paraguay	128	507	142
VOL030	Paraguay030	Paraguay	128	507	142
VOL031	Paraguay031	Paraguay	128	507	142
VOL032	Paraguay032	Paraguay	128	507	142
VOL033	Paraguay033	Paraguay	128	507	142
VOL034	Paraguay034	Paraguay	128	507	142
VOL035	Paraguay035	Paraguay	128	507	142
VOL036	Paraguay036	Paraguay	128	507	142
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VOL039	Paraguay039	Paraguay	128	507	142
VOL040	Paraguay040	Paraguay	128	507	142
VOL041	Paraguay041	Paraguay	128	507	142
VOL042	Paraguay042	Paraguay	128	507	142
VOL043	Paraguay043	Paraguay	128	507	142
VOL044	Paraguay044	Paraguay	128	507	142
VOL045	Paraguay045	Paraguay	128	507	142
VOL046	Paraguay046	Paraguay	128	507	142
VOL047	Paraguay047	Paraguay	128	507	142
VOL048	Paraguay048	Paraguay	128	507	142
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VOL050	Paraguay050	Paraguay	128	507	142

metadata

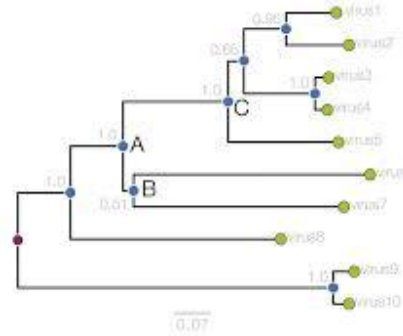
Internal

Genomic Epidemiology Challenges

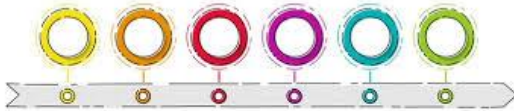
DATA EXPLORATION



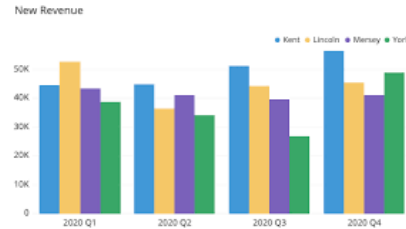
map



phylogeny



timeline



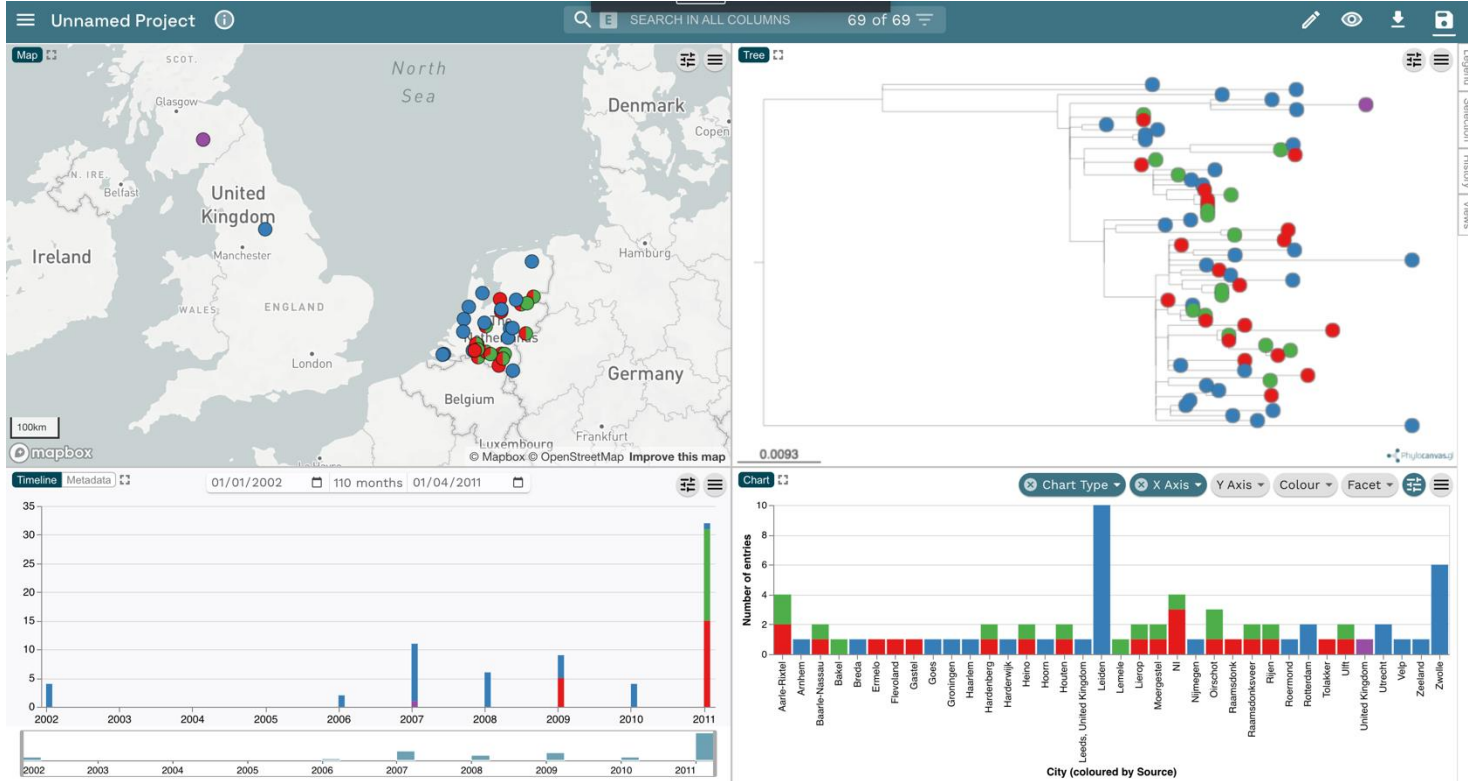
analysis

Internal

Genomic Epidemiology Challenges

DATA EXPLORATION

Microreact





wellcome
connecting
science

Demo Time!



Microreact

Genomic Epidemiology Challenges

BIOINFORMATICS ACCESSIBILITY

RAW DATA

fastqs

[illegible]

Genomic Epidemiology Challenges

BIOINFORMATICS ACCESSIBILITY

RAW DATA

fastqs

[illegible]

QC & assembly



ASSEMBLY

fasta

```
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Genomic Epidemiology Challenges

BIOINFORMATICS ACCESSIBILITY

RAW DATA

fastqs

[illegible]

ASSEMBLY

fasta

>NODE_1_length_1901424_cov_73.350607
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 GTTAAGTACTTGACATCTGGGAGCTAAGCTTGTGATTCCTCTCTCGAAGAACCTCAAG
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DOWNSTREAM ANALYSIS

- Typing
- Contextualization
- AMR marker identification
- Storage



Pathogenwatch

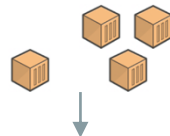
<http://pathogen.watch>

- Fast predictions of resistant genotypes and relatedness from whole genome assemblies
- Contextualization with public data
- Real-time analytics and genomic epidemiology
- User-friendly web application

Sequences and Metadata
(Fastq assembled)



Speciation



Species-specific analytics

Analytics
(containers)



Available Community-Developed Analytics



athogenwatch

	Species ID (Mash+RefSeq)	MLST	Additional Genotyping	Serotyping	cgMLST Network View	Core SNP trees Collection View	Curated AMR prediction	Plasmid Inc types (plasmidFinder)	Curated virulence prediction	
<i>A. baumannii</i>	Kleborate			Kaptive					Kleborate	
<i>P. aeruginosa</i>										
<i>K. pneumoniae</i> complex				Kaptive						
<i>E. coli</i>										
<i>Shigella</i> spp.										
<i>E. cloacae</i> complex										
<i>E. faecium</i>										
<i>S. aureus</i>										
<i>C. coli/jejuni</i>							Ciprofloxacin			
<i>S. enterica</i>				SISTR						
<i>S. Typhi</i>			GenoTyphi	SISTR						
<i>N. gonorrhoeae</i>			NG-MAST/STAR							
<i>S. pneumoniae</i>			GPSCs, PBPs	SeroBA						
<i>M. tuberculosis</i>										
<i>V. cholerae</i>										
<i>L. monocytogenes</i>										
<i>S. equi</i>										
<i>R. salmoninarum</i>										
<i>C. auris</i>										
Zika										
SARS-CoV-2			Pangolin, Mutations							

Internal

Demo Time!



How is this computationally possible?



How is this computationally possible?

Epicollect compression algorithm

$$r : \rho \left(\frac{\partial u_r}{\partial t} + u_r \frac{\partial u_r}{\partial r} + \frac{u_\phi}{r} \frac{\partial u_r}{\partial \phi} + u_z \frac{\partial u_r}{\partial z} - \frac{u_\phi^2}{r} \right) = - \frac{\partial p}{\partial r} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_r}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_r}{\partial \phi^2} + \frac{\partial^2 u_r}{\partial z^2} - \frac{u_r}{r^2} - \frac{2}{r^2} \frac{\partial u_\phi}{\partial \phi} \right] + \rho g_r$$

Data-flo GPU acceleration

$$\phi : \rho \left(\frac{\partial u_\phi}{\partial t} + u_r \frac{\partial u_\phi}{\partial r} + \frac{u_\phi}{r} \frac{\partial u_\phi}{\partial \phi} + u_z \frac{\partial u_\phi}{\partial z} + \frac{u_r u_\phi}{r} \right) = - \frac{1}{r} \frac{\partial p}{\partial \phi} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_\phi}{\partial \phi^2} + \frac{\partial^2 u_\phi}{\partial z^2} + \frac{2}{r^2} \frac{\partial u_r}{\partial \phi} - \frac{u_\phi}{r^2} \right] + \rho g_\phi$$

Microreact quantum demultiplex

$$z : \rho \left(\frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + \frac{u_\phi}{r} \frac{\partial u_z}{\partial \phi} + u_z \frac{\partial u_z}{\partial z} \right) = - \frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_z}{\partial \phi^2} + \frac{\partial^2 u_z}{\partial z^2} \right] + \rho g_z.$$